8.03SC Problem Set 3

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Problem 1

Here we consider a double pendulum, each with one degree of freedom, as shown in the figure above. Mass M_1 is connected by a massless rigid rod of length L to a fixed origin. Its x-coordinate is X_1 and it makes an angle θ_1 with respect to the vertical (y-axis). Mass M_2 is connected by a massless rigid rod of length L to mass M_1. Its x-coordinate is X_2 and it makes an angle θ_2 with respect to the vertical direction.

You may assume that the 'hinges' at the origin and on mass M_1 are frictionless. Gravity points downward in the y-direction. You may assume that all oscillations are small, i.e., that θ_1 and θ_2 are small and terms of order $O(\theta^2)$ are negligible.

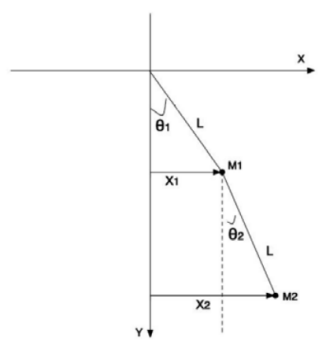

Figure 1: Two circuits

a. Write down the equations of motion for the two oscillating masses.

b. Determine the two normal mode frequencies of oscillation in the small amplitude limit as a function of ω_0, and α, where $\omega_0^2 = \frac{g}{L}$ and $\alpha = \frac{M_2}{M_1}$. (You don't have to solve the corresponding amplitude ratios.)

c. What are the two normal mode frequencies if $\alpha \to \infty$? Do your results make sense?

d. What are the two normal mode frequencies if $\alpha \to 0$? Do your results make sense?

a.

$\theta_1, \theta_2 \ll 1$

Motion of M_2 can be represented as

$$M_2\ddot{X}_2 = -M_2 g\theta_2 \implies \ddot{X}_2 = -g\theta_2$$

Let Tension in the String connecting the 2 Masses be T. So,

$$T \approx M_2 g$$

So, Motion of M_1 can be represented as

$$M_1\ddot{X}_1 = -M_1 g\theta_1 + T\theta_2 = -M_1 g\theta_1 + M_2 g\theta_2$$

By Geometry, we can say that

$$X_1 = L\theta_1 \implies \theta_1 = \frac{X_1}{L}$$

$$X_2 = X_1 + L\theta_2 \implies \theta_2 = \frac{X_2 - X_1}{L}$$

This gives the Equations of Motion to be

$$\ddot{X}_1 = -\Big(\frac{g}{L} + \frac{M_2}{M_1}\frac{g}{L}\Big)X_1 - \Big(-\frac{M_2}{M_1}\frac{g}{L}\Big)X_2$$

$$\ddot{X}_2 = -\Big(-\frac{g}{L}\Big)X_1 - \Big(\frac{g}{L}\Big)X_2$$

b.

With $\omega_0^2 = \frac{g}{L}$ and $\alpha = \frac{M_2}{M_1}$, Our equation of turns into

$$\ddot{X}_1 = -(\omega_0^2 + \alpha\omega_0^2)X_1 - (-\alpha\omega_0^2)X_2$$

$$\ddot{X}_2 = -(-\omega_0^2)X_1 - (\omega_0^2)X_2$$

Let Normal Mode Frequency be ω
Characteristic Equation comes out to be

$$\begin{vmatrix} (\omega_0^2 + \alpha\omega_0^2 - \omega^2) & (-\alpha\omega_0^2) \\ (-\omega_0^2) & (\omega_0^2 - \omega^2) \end{vmatrix} = 0$$

$$\implies \omega^2 = \Big(\frac{(2+\alpha) \pm \sqrt{\alpha^2 + 4\alpha}}{2}\Big)\omega_0^2$$

Note that $(2+\alpha) > \sqrt{\alpha^2 + 4\alpha}$
Therefore the Two Normal Mode Frequencies are

$$\omega_1 = \left(\sqrt{\frac{(2+\alpha) - \sqrt{\alpha^2 + 4\alpha}}{2}}\right)\omega_0 \textbf{ and } \omega_2 = \left(\sqrt{\frac{(2+\alpha) + \sqrt{\alpha^2 + 4\alpha}}{2}}\right)\omega_0$$

c.

$$\alpha \to \infty \implies \omega_1 \to 0 \textbf{ and } \omega_2 \to \infty$$

These results make physical sense.

In the limit $\alpha \to \infty$ i.e. $M_2 \gg M_1$, one of the normal–mode frequencies tends to zero, while the other tends to infinity. The low–frequency mode corresponds to a *soft mode* in which the system can move with almost no restoring force, leading to very slow motion. The high–frequency mode corresponds to an infinitely stiff deformation, requiring an infinite restoring force and therefore not accessible to any finite–energy motion.

Thus, as the coupling becomes very strong, one mode becomes effectively frozen (removed from the dynamics), while the remaining mode governs the physical motion of the system. This behavior is exactly what one expects when a constraint is imposed, and therefore the results are physically reasonable.

d.

$$\alpha \to 0 \implies \omega_1 = \omega_2 = 1$$

In the limit $\alpha \to 0$, the coupling between the two masses vanishes and the system reduces to two independent simple pendulums. Each mass therefore oscillates with its natural frequency ω_0, and the two normal–mode frequencies become equal. This degeneracy reflects the absence of coupling and the fact that the motions of the two masses are independent in this limit.

Problem 2

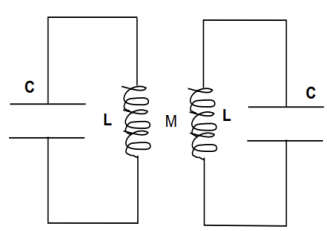

Figure 2: Two circuits

Consider two identical LC circuits as shown in Figure 2. The two inductors are brought close together such that their mutual inductance M results in a coupling between the currents flowing in the two circuits.

a. Find the frequencies of normal modes as a function of given parameters.

b. What current patterns correspond to these normal modes? Could you use symmetry arguments to discover these modes?

a.

Let components in the left be C_1 and L_1 and on the right be C_2 and L_2. Let charge moving along the left be q_1 and along the right be q_2. We will assume that in both the branches, the Current flows in Anti-Clockwise Direction in both the Circuits.

Therefore, for the K.V.L. Equations becomes

$$\frac{q_1}{C} + L\frac{d}{dt}\frac{dq_1}{dt} - M\frac{d}{dt}\frac{dq_2}{dt} = 0 \implies \frac{q_1}{C} + L\frac{d^2q_1}{dt^2} - M\frac{d^2q_2}{dt^2} = 0$$

$$\frac{q_2}{C} + L\frac{d}{dt}\frac{dq_2}{dt} - M\frac{d}{dt}\frac{dq_1}{dt} = 0 \implies -M\frac{d^2q_1}{dt^2} + \frac{q_2}{C} + L\frac{d^2q_2}{dt^2} = 0$$

Let

$q_1 = A_1\mathbf{Re}[e^{i\omega\, t}]$ and $q_2 = A_2\mathbf{Re}[e^{i\omega\, t}]$ here ω represents Normal Mode Frequency

This enables us to write the K.V.L. equations in the Matrix form as

$$\begin{bmatrix} \left(\frac{1}{C} - \omega^2 L\right) & (M\omega^2) \\ (M\omega^2) & \left(\frac{1}{C} - \omega^2 L\right) \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\implies \begin{vmatrix} \left(\frac{1}{C} - \omega^2 L\right) & (M\omega^2) \\ (M\omega^2) & \left(\frac{1}{C} - \omega^2 L\right) \end{vmatrix} = 0$$

$$\implies \omega^2 = \frac{1}{C(L \pm M)}$$

Therefore

Normal Modes Angular Frequencies :- $\sqrt{\frac{1}{C(L-M)}}$ **&** $\sqrt{\frac{1}{C(L+M)}}$

b.

$$\omega^2 = \frac{1}{C(L \pm M)}$$

Mode 1: $\omega^2 = \dfrac{1}{C(L-M)}$

Substituting,

$$\frac{1}{C} - \omega^2 L = \frac{1}{C} - \frac{L}{C(L-M)} = -\frac{M}{C(L-M)}, \qquad M\omega^2 = \frac{M}{C(L-M)}.$$

Thus the matrix equation becomes

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = 0,$$

which gives

$$A_1 = A_2.$$

This mode corresponds to the two charges oscillating *in phase*.

Mode 2: $\omega^2 = \dfrac{1}{C(L+M)}$

Substituting,

$$\frac{1}{C} - \omega^2 L = \frac{1}{C} - \frac{L}{C(L+M)} = \frac{M}{C(L+M)}, \qquad M\omega^2 = \frac{M}{C(L+M)}.$$

The matrix equation becomes

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = 0,$$

which gives

$$A_1 = -A_2.$$

This mode corresponds to the two charges oscillating *out of phase*.

The normal modes are:

- **In–phase mode:** $A_1 = A_2$, $\quad \omega = \sqrt{\dfrac{1}{C(L-M)}}$
- **Out–of–phase mode:** $A_1 = -A_2$, $\quad \omega = \sqrt{\dfrac{1}{C(L+M)}}$

Problem 3

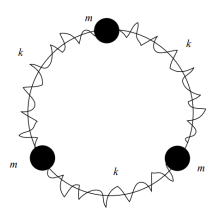

Figure 3: Three Masses on Circle

Consider three identical masses constrained to move on a frictionless circle. The masses are connected with identical springs each with spring constant k (see Figure 3). The circle is large such that you can ignore any effects related to the curvature. The circle is horizontal such that gravity can be ignored.

a. Find equations of motion for the three masses in terms of the small displacements from the equilibrium position of each mass.

b. Determine the frequencies and the relative amplitudes for each of the normal modes. Make a simple sketch of the motion of the masses for each of the normal modes. How many different frequencies are there in this system?

c. Because the masses are connected in a circle some of the results of normal mode calculations do not correspond to oscillatory motion. Explain why.

a.

Let the masses be labeled as m_1, m_2 and m_3 arranged in Anti-Clockwise fashion and each gets displaced by x_1, x_2 and x_3 respectively in Anti-Clockwise Direction.

Using Geometry and Symmetry Equation of Motions comes out to be

$$m_1\ddot{x}_1 = k((x_2 - x_1) - (x_1 - x_3))$$

$$m_2\ddot{x}_2 = k((x_3 - x_2) - (x_2 - x_1))$$

$$m_3\ddot{x}_3 = k((x_1 - x_3) - (x_3 - x_2))$$

In the Question it is given $m_1 = m_2 = m_3 = m$
This gives the Equations to be

$$\ddot{x}_1 = -\frac{2k}{m}x_1 + \frac{k}{m}x_2 + \frac{k}{m}x_3$$

$$\ddot{x}_2 = \frac{k}{m}x_1 - \frac{2k}{m}x_2 + \frac{k}{m}x_3$$

$$\ddot{x}_3 = \frac{k}{m}x_1 + \frac{k}{m}x_2 - \frac{2k}{m}x_3$$

In Matrix Form :

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}'' = \begin{bmatrix} -\frac{2k}{m} & \frac{k}{m} & \frac{k}{m} \\ \frac{k}{m} & -\frac{2k}{m} & \frac{k}{m} \\ \frac{k}{m} & \frac{k}{m} & -\frac{2k}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

b.

Let

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} \mathbf{Re}\{e^{i\omega}\} \text{ here } \omega \text{ is Normal Mode Frequency}$$

So the Characteristic Equation comes out to be

$$\begin{vmatrix} \frac{2k}{m} - \omega^2 & -\frac{k}{m} & -\frac{k}{m} \\ -\frac{k}{m} & \frac{2k}{m} - \omega^2 & -\frac{k}{m} \\ -\frac{k}{m} & -\frac{k}{m} & \frac{2k}{m} - \omega^2 \end{vmatrix} = 0$$

$$\implies \omega^2 = 0 \text{ or } \frac{3k}{m}$$

Relative Amplitudes :-
$\omega^2 = 0$

$$\frac{k}{m}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\implies A_1 = A_2 = A_3$$

$\omega^2 = \frac{3k}{m}$

$$\frac{k}{m}\begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix}\begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\implies A_1 + A_2 + A_3 = 0$$

Interpretation of Normal Modes

Zero–frequency mode ($\omega = 0$)
For $\omega^2 = 0$, the relative amplitudes satisfy

$$A_1 = A_2 = A_3.$$

Thus, all three masses move together with equal displacement and phase. Since the springs are not stretched in this motion, there is no restoring force. This mode corresponds to a rigid translation of the entire system around the circle and therefore does not represent oscillatory motion.

Oscillatory modes $\left(\omega = \sqrt{\frac{3k}{m}}\right)$
For $\omega^2 = \frac{3k}{m}$, the amplitudes satisfy

$$A_1 + A_2 + A_3 = 0.$$

This condition admits two linearly independent solutions, indicating that the frequency

$$\omega = \sqrt{\frac{3k}{m}}$$

is doubly degenerate. These modes correspond to oscillatory motion in which the masses move relative to one another while maintaining zero net displacement, resulting in periodic stretching and compression of the springs.

Summary :-
The system has:

- One non-oscillatory normal mode with $\omega = 0$, corresponding to rigid motion around the circle.
- Two degenerate oscillatory normal modes with frequency $\omega = \sqrt{\frac{3k}{m}}$.

Hence, there is only one nonzero oscillation frequency in this system, but it occurs with degeneracy due to the rotational symmetry of the configuration.

c.

One of the normal modes has zero angular frequency, $\omega = 0$, and therefore does not correspond to oscillatory motion. This occurs because the system is invariant under uniform translation of all three masses around the circle.

In this mode the relative amplitudes satisfy

$$A_1 = A_2 = A_3,$$

so all masses move together with the same displacement. As a result, the springs are neither stretched nor compressed, and no restoring forces act on the masses.

Since oscillatory motion requires a restoring force, this mode represents rigid motion of the entire system rather than an oscillation. The appearance of a zero-frequency normal mode is therefore a direct consequence of the circular symmetry of the system.

Problem 5

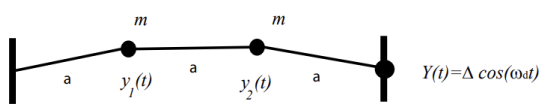

Figure 5: Two Beads

Two identical beads, each of mass m, are equally spaced along a massless string of length $3a$ (see Figure 5). Consider the system to be on a frictionless horizontal surface. Initially both ends of the string are fixed ($\Delta = 0$). Assume that the string is under tension T at all times. Beads can execute small amplitude oscillations perpendicular to the string (displacements are exaggerated in the Figure!).

a. Find equations of motion for the two beads in terms of displacement from the equilibrium $y_1(t)$, $y_2(t)$.

b. Find and sketch the motion of the normal modes and calculate the normal mode frequencies for the system.

Assume now that the rightmost attachment point undergoes harmonic oscillation $y(t) = \Delta\cos(\omega_d t)$.

c. Find the steady state amplitude of the motion of the two masses as a function of the driving frequency ω_d and the amplitude Δ.

a.

By Geometry, the downward Force on the left bead is

$$T(\frac{y_1}{a}) + T(\frac{y_1 - y_2}{a}) = \frac{T}{a}(2y_1 - y_2)$$

Similarly, downward Force on the right bead is

$$T(\frac{y_2 - y_1}{a}) + T(\frac{y_2 - Y}{a}) = \frac{T}{a}(2y_2 - y_1 - Y)$$

Therefore, the Equations of Motion comes out to be

$$m\ddot{y}_1 = \frac{T}{a}(2y_1 - y_2)$$

$$m\ddot{y}_2 = \frac{T}{a}(-y_1 + 2y_2 - Y)$$

b.

For Normal Modes, setting $Y = 0$

The equation of Motion becomes

$$\ddot{y}_1 = \frac{T}{ma}(2y_1 - y_2) \text{ and } \ddot{y}_2 = \frac{T}{ma}(-y_1 + 2y_2)$$

$$\implies \frac{d^2}{dt^2}\left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = \begin{bmatrix} \frac{2T}{ma} & -\frac{T}{ma} \\ -\frac{T}{ma} & \frac{2T}{ma} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

If ω is the Normal Mode Frequency and

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \mathbf{Re}(e^{i\omega t})$$

The Characteristic Equation comes out to be

$$\begin{vmatrix} \frac{2T}{ma} - \omega^2 & -\frac{T}{ma} \\ -\frac{T}{ma} & \frac{2T}{ma} - \omega^2 \end{vmatrix} = 0$$

$$\implies \omega^2 = \frac{T(2 \pm 1)}{ma}$$

Relative Amplitude :-

$\omega^2 = \frac{T}{ma}$

$$\frac{T}{ma}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = 0$$

$$\implies A_1 = A_2$$

$\omega^2 = \frac{3T}{ma}$

$$\frac{3T}{ma}\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = 0$$

$$\implies A_1 = -A_2$$

Physical Interpretation of the Normal Modes

Mode 1: $\omega^2 = \dfrac{T}{ma}$ (in–phase mode)

For this mode, the relative amplitudes satisfy

$$A_1 = A_2.$$

Thus both beads move with the same displacement and in the same phase. The string segment between the two beads does not change its length, and therefore does not contribute to the restoring force. Only the two outer string segments are stretched. As a result, the effective restoring force is smaller, leading to the lower normal–mode frequency.

Mode 2: $\omega^2 = \dfrac{3T}{ma}$ (out–of–phase mode)
For this mode, the relative amplitudes satisfy

$$A_1 = -A_2.$$

The two beads move with equal amplitude but in opposite directions. In this motion, the middle string segment is stretched and compressed maximally, in addition to the outer segments. Consequently, the effective restoring force is larger, giving rise to the higher normal–mode frequency.

In summary, the lower–frequency normal mode corresponds to symmetric (in–phase) motion with weaker restoring forces, while the higher–frequency normal mode corresponds to antisymmetric (out–of–phase) motion with stronger restoring forces. This ordering of frequencies is physically consistent with the number of string segments that are deformed in each mode.

c.

The general motion can be written as a superposition of normal modes:

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = Q_1(t) \begin{bmatrix} 1 \\ 1 \end{bmatrix} + Q_2(t) \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

where $Q_1(t)$ and $Q_2(t)$ are the normal–mode coordinates describing the time–dependent amplitudes of the in–phase and out–of–phase modes, respectively.
This gives us

$$y_1 = Q_1 + Q_2 \text{ and } y_2 = Q_1 - Q_2$$

Therefore,

$$\ddot{Q}_1 + \ddot{Q}_2 = \frac{T}{ma}(2(Q_1 + Q_2) - (Q_1 - Q_2)) = \frac{T}{ma}(Q_1 + 3Q_2)$$

$$\ddot{Q}_1 - \ddot{Q}_2 = \frac{T}{ma}(-(Q_1 + Q_2) + 2(Q_1 - Q_2) - Y) = \frac{T}{ma}(Q_1 - 3Q_2 - Y)$$

This gives us

$$\ddot{Q}_1 = \frac{T}{ma}(Q_1 - \frac{Y}{2}) \text{ \& } \ddot{Q}_2 = \frac{T}{ma}(3Q_2 + \frac{Y}{2})$$

$$\implies \ddot{Q}_1 - \frac{T}{ma}Q_1 = -\frac{T\Delta}{2ma}\cos(\omega_d t) \text{ \& } \ddot{Q}_2 - \frac{3T}{ma}Q_2 = \frac{T\Delta}{2ma}\cos(\omega_d t)$$

Now the question implicitly asked to find the Steady-State solutions. So we will find the Particular Solutions.

This Gives us

$$Q_{1P} = \frac{T\Delta}{2\omega_d(ma\omega_d + T)}\cos(\omega_d t) \ \& \ Q_{2P} = \frac{-T\Delta}{2\omega_d(ma\omega_d + 3T)}\cos(\omega_d t)$$

Therefore the Steady State Solution is given by

$$y_{1P} = \left(\frac{T\Delta}{2\omega_d(ma\omega_d + T)} - \frac{T\Delta}{2\omega_d(ma\omega_d + 3T)}\right)\cos(\omega_d t)$$

$$y_{2P} = \left(\frac{T\Delta}{2\omega_d(ma\omega_d + T)} + \frac{T\Delta}{2\omega_d(ma\omega_d + 3T)}\right)\cos(\omega_d t)$$

Hence,

Steady-State Amplitude :-

$$y_{1P} = \left(\frac{T\Delta}{2\omega_d(ma\omega_d + T)} - \frac{T\Delta}{2\omega_d(ma\omega_d + 3T)}\right)$$

$$y_{2P} = \left(\frac{T\Delta}{2\omega_d(ma\omega_d + T)} + \frac{T\Delta}{2\omega_d(ma\omega_d + 3T)}\right)$$